MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Customer: Mansourah Massarah Gold Project
MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: GUIDELINES TOPOGRAPHICAL SURVEY

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TABLE OF CONTENTS

1. INTRODUCTION

1.1 PURPOSE……..3

1.2 KEY TERMS AND DEFINITION……3

2. BACKGROUND

2.1 HORIZONTAL DATUM……4

2.2 VERTICAL DATUM……5

2.3 DISTANCE ACCURACY STANDARDS……5

2.4 UNITS……..5

3. DATUM TRANSFORMATION

3.1 BASIS OF TRANSFORMATION……5

3.2 BENCHMARK……6

4 PRCOEDURE TO BE ADOPTED FOR SITE WORKS

4.1 EXECUTION……6

4.2 INSTRUMENT USED……7

5 PROGRESS REPORTING……8

6 SUBMISSION OF DOCUMENTATION

6.1 SURVEY REPORT……...8

7 REFERENCE……..9

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1.0 Introduction

This procedure has been developed to provide guidelines for all survey set out and control on Project.

Due to the mix of horizontal and vertical datum's on the project, it is essential this manual be strictly followed to avoid design/construction errors.

1.1 Purpose

The purpose of this Specification is to:

- Identify applicable Standards and guidelines to be used for the Works
- Set the minimum quality control standards and tolerances applicable
- Define the general requirements for the Works including safety and permitting, and
- List the typical deliverables expected upon completion of the Works.

1.2 Key Terms And Definitions

AASHTO American Association of State Highway and Transportation Officials.

Contractor An entity, engaged or approved by the Owner, entrusted with a Contract to perform Work or cause Work to be performed.

Employer Saudi Arabian Mining Company (Ma'aden) and its authorized Representatives and respective affiliates.

FGC Federal Geodetic Control Committee.

MSL Mean Sea Level

SI International system of units.

Sit Geographical location where Services are to be performed.

UTM Universal Transverse Mercator

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2. Background

As Project Mansoura Massarah Gold Project must interface with existing infrastructure in the MMGP facility and the overall sites. A brief outline of the background to these multiple datum's follows.

2.1 Horizontal Datum

WGS84 (UTM Zone 38 Coordinates) – legal definition of the site boundary location, and interfaces with external utilities/clients.
The following benchmarks will be jointly verified with Client or Client's Consultant before starting the survey work.

BENCH MARK NUMBER	NORTHING	EASTING	ELEVATION
BM-14	2525710.871	262080.999	940.596
BM-15	2525197.156	262647.441	938.043

The above reference benchmarks are established near the Pressure reducing station for the Ma'aden Gold project - Adduwayhi Junction. Additional benchmarks have to be established taking reference from these bench mark. The location and values for the primary benchmarks are as per ITB (Reference doc no: MD-212-0110-EG-CV-DEL-0001).

The accuracy of horizontal measurements for establishing primary horizontal control shall meet the requirements of a third-order, class I survey, as defined by FGCC Standards (1984), Table 2.1, unless specified otherwise. Secondary horizontal control networks shall be closed and traverses shall be tied to primary horizontal controls. The accuracy of controls shall meet the requirements of a third-order, class II survey, as defined by FGCC Standards (1984), Section 2.1, unless specified otherwise.

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2.2 Vertical Datums

Users must note that WGS84 benchmark values provided by MMGP Shall be used. The accuracy of vertical measurements to establish primary vertical control shall meet the Requirements of a second-order, class II survey, as defined by FGCC Standards (1984), Table 2.2, unless specified otherwise. Secondary vertical control networks shall be level loops tied to primary vertical controls. The accuracy of controls shall meet the requirements of a third-order, class II survey, as defined by FGCC Standards (1984), Section 2.2, unless specified.

2.3 Distance Accuracy Standards

Classification	Minimum distance accuracy
3rd order class 1	1:10000

2.4 Units

All measurements shall be in SI Units. Length shall be measured in meters (m) correct to two decimal places. Areas shall be worked out in square meters (sqm) and volume in cubic meters(cum) rounded off to two decimal places.

3.0 Datum Transformations

3.1 Basis of Transformation

Transformations between the local plant grid systems used for design and construction set out and the global UTM grid system are necessary in order to ensure accuracy where external infrastructure interfaces with the Project MMGP SITE.

As the project site is approximately 25 km away from the existing bench mark, every 2 km secondary bench mark will be created from the reference of existing BM-14 and BM-15. Following the similar procedure, permanent benchmark at the MMGP project site will be established.

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3.2 Benchmark

Benchmarks shall be located in strategic locations, protected from traffic and constructed as per the directions of the Employer. Benchmarks shall be clearly engraved with the reduced level and co-ordinates with respect to the survey grid.

Overall Planning

- Plan the campaign carefully
- Consider the job, number of points, accuracy needed
- Consider connection to existing control
- Consider the transformation to local coordinates
- Consider the best ways to observe and compute
- For high accuracy, keep baselines as short as possible
- Use reference stations
  - Consider the need for independent checks:
  - Occupying points twice in different windows
  - Closing traverse loops
- Measuring independent baselines between points
- For high-accuracy work, try not to squeeze the maximum number of points into a window

4.0 Procedures to be adopted for site works

4.1 Execution

The Work at Site shall be carried out under full time supervision by a qualified Engineer or Surveyor.

The Engineer or Surveyor shall be responsible for and capable of coordinating the work of the surveying teams, setting out the work accurately, identifying the type of instruments to be deployed and methodology of surveying to achieve the requirements of the Specification.

However **Ma'aden to confirm the reference bench mark points to L&T**. Appropriate instruments should be deployed to achieve the specified accuracy of the work. Proper

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precautions for avoiding graduation errors, instrumental and personal errors should be scrupulously observed.

4.2 INSTRUMENT USED

Trimble R4S RTK base & rover and level instrument will be used and all instructions Will be followed as per the Instrument Operation Manual.

Main Plant will have 02 Master Benchmarks at a distance of 200 meters. These will be secured and protected throughout of the construction of project and no intermediate obstruction will be permitted.

Positions both in plan and elevation of all natural and artificial features of the area such as waterways, trees, cultivation, fences, and roads including culverts and crossings, foot tracks, other permanent objects like telephone posts and transmission towers etc. will be captured and subsequently shown on survey maps.

All earth deposits, depressions, hills and valleys within the area(s) will be surveyed and plotted on maps by contours. Where grid establishment is not possible e.g. in the case of steep slopes and densely vegetated areas, the method of survey, contour intervals etc shall be as directed by the Ma'aden. Any unusual conditions or formations on the ground such as rock outcrops, springs/falls etc. will also be surveyed accordingly to facilitate plotting on the maps.

The field work shall include the following:

- Trimble R4S GPS method to be used static observation for Benchmark.
- Establishing horizontal and vertical controls
- Locating reference grids and bench marks in the area
- Surveying spot levels and plotting contours, and
- Locating and plotting natural and manmade features as described above.

The grid for the survey work shall be established in north-south & east-west direction corresponding to magnetic north or the plant north as directed by Ma'aden. Clearance and cutting of trees as required to facilitate the survey work also forms part of the Work.

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5 Progress Reporting

progress report will be prepared indicating the status of the Work, methodology adopted for surveying, instruments deployed, staff working on the Site and difficulties encountered during execution of the Work etc.

6 Submission of Documentation

Hardcopies and softcopies of all documentation as per the contract will be submitted not limited to

- survey report
- survey maps
- field notes
- calculations.
- Photograph

Survey maps shall also be provided in electronic format. Field data shall be submitted from time to time if requested, and as per progress of the work.

Survey maps of the Site will be submitted in 1:1000 scale with grid lines on 50m interval and contour interval of 0.5 m in AutoCAD or micro station 3D format. One hard copy of final drawing will be submitted along with CD.

Overall plant and facility drawings and maps shall show a minimum of three (3) survey monuments with UTM coordinates and MSL elevations. All documents shall be submitted for review and approval.

All documentation submitted shall generally be compatible with the following software systems, or file formats,

- Microsoft Office suite
- .dtm, .dxf,. dgn, .dwg, .CSV
- .pdf.

6.1 Survey Report

The report shall give introduction to the Site, methodology adopted for surveying the areas, calculation of errors, transfer of bench marks and any other calculations required for

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surveying and preparation of the survey maps. Details of trees with their name, numbers and girths shall also form part of the survey report.

The survey report will also cover the following:

- General Site observation such as location of existing structures, installations, access
- Roads, rivers and irrigation canals etc.
- Presence of any wells and/or tube wells in the Site indicating their water level also
- Any rock outcrops on the Site, and
- Existing drainage pattern of the Site, possibility of water logging and high flood level of the area.

7 Reference

Maaden Engineering Manual for Topographical Survey # MD-101-SMEM-EG-CV-SPC-0008

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